

SemanticSplat: Graph-Pruned Semantic Search

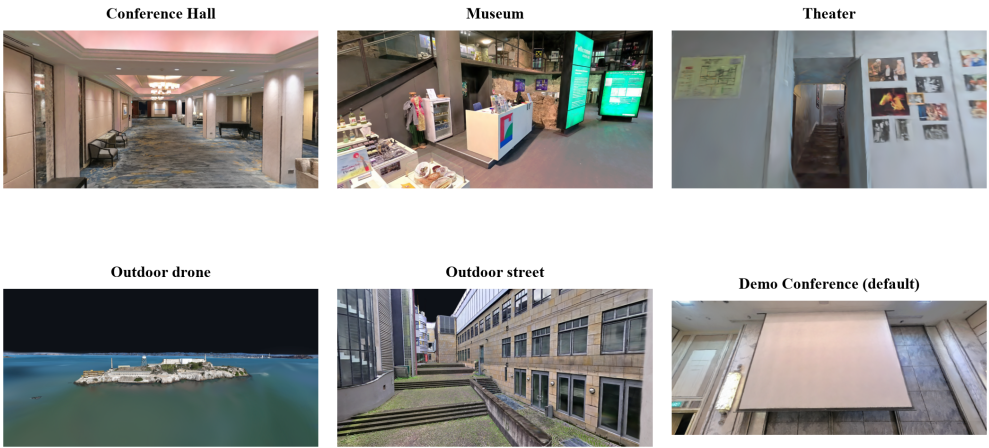
Current Status and Research Roadmap

Technical report | 15 July 2026 | Branch: codex/project-site-method-figure | Evidence base: 46ac2bf

Bottom line. The team has completed a reproducible semantic-search prototype, official Replica and ScanNet oracle-map pilots, a five-scene ConceptGraphs run, baseline smoke evidence, and an anonymous ECCV-format workshop manuscript. The remaining work is concentrated in submission actions and stronger non-oracle, same-protocol science rather than basic project plumbing.

97 captured views	150 internal queries	575 Replica GT boxes	392 ScanNet GT boxes
144 passing tests	42/42 paper number checks	11 paper pages incl. refs	13 cited related works

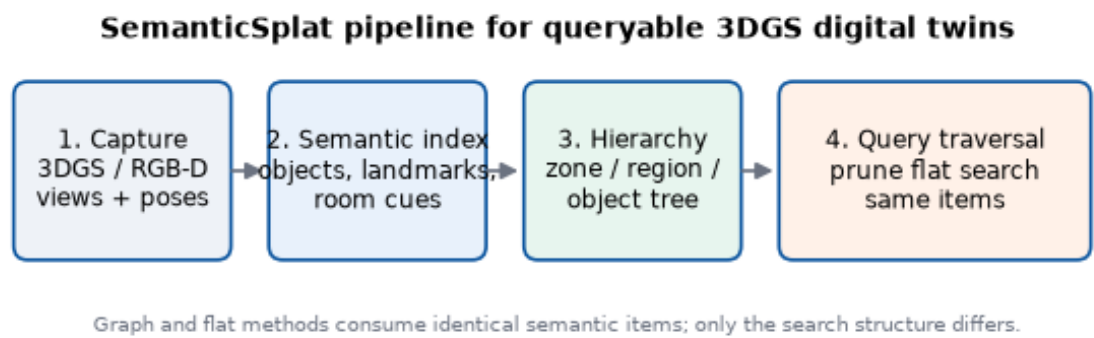
Digital-twin pilot scenes (manual captures)



Five manually captured pilot scenes plus the default qualitative view used for demos.

What we completed: a reproducible evaluation prototype

The current system converts captured or imported scene observations into a semantic index and hierarchy, runs graph-pruned or flat query variants, and saves canonical per-query results for evaluation. Construction cost is separated from query-time cost.



System flow. Graph and flat search consume the same semantic items; only the search structure differs.

Completed work and saved evidence

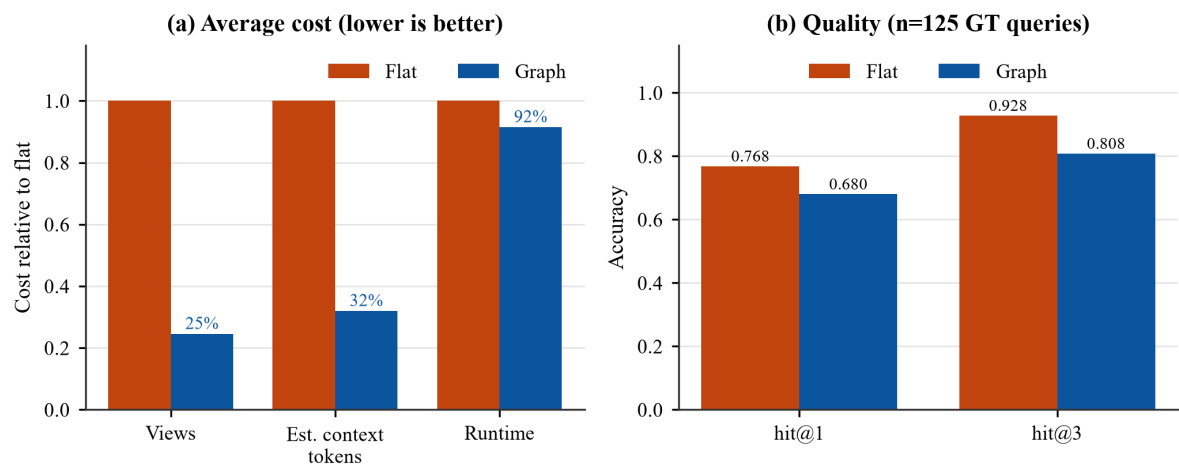
Area	Completed result	Evidence
Captured data	5 scenes, 97 RGB-D views, 1,066 semantic items	backend/data/scenes/
Internal GT	150 queries; 125 carry verified expected view IDs	docs/benchmarks/benchmark_queries_v2.json
Public data	8 Replica + 8 ScanNet scenes with official 3D object boxes	backend/data/scenes/replica_* and scannet_*
Evaluation	Graph, graph+fallback, flat lexical, flat embedding, and ablations	outputs/public_datasets/
External baselines	ConceptGraphs five-scene run; LangSplat official-sofa smoke	outputs/baselines/
Article	Anonymous ECCV 2026 review PDF, 10 content pages + references	papers/twinworld/main.pdf
Verification	144 tests and a 42-check paper evidence gate	tests/ and twinworld_number_check.md

Claim boundary: this is a semantic retrieval and reasoning prototype. Public pilots search official semantic maps; they do not measure automatic semantic perception.

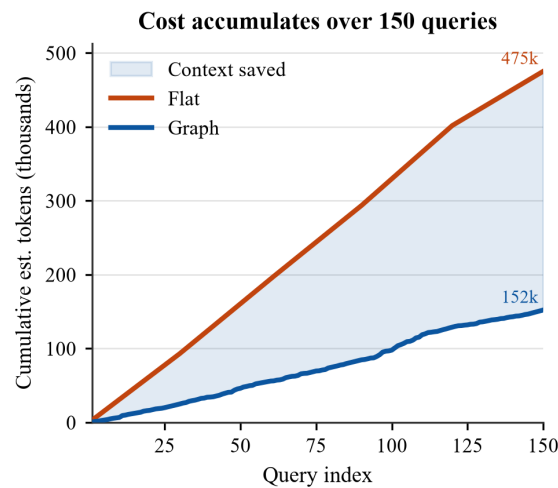
Hierarchy cuts context sharply, with a measured quality cost

Across 150 same-input queries, graph traversal checks 75.5% fewer views and uses 68.2% fewer estimated context tokens. On the 125 queries with verified view GT, however, graph hit@1 and hit@3 remain below flat lexical search. This is a cost-quality frontier, not a claim of free quality preservation.

Metric	Flat	Graph	Difference / interpretation
Views checked (mean)	19.40	4.77	75.5% fewer
Est. context tokens (mean)	3168.2	1014.3	68.2% fewer
Context characters (mean)	12701.4	4084.8	Less serialized evidence
hit@1 (n=125)	0.768	0.680	Graph -0.088
hit@3 (n=125)	0.928	0.808	Graph -0.120



Normalized cost and verified-view hit@k. The graph saves context but currently trails flat lexical retrieval quality.



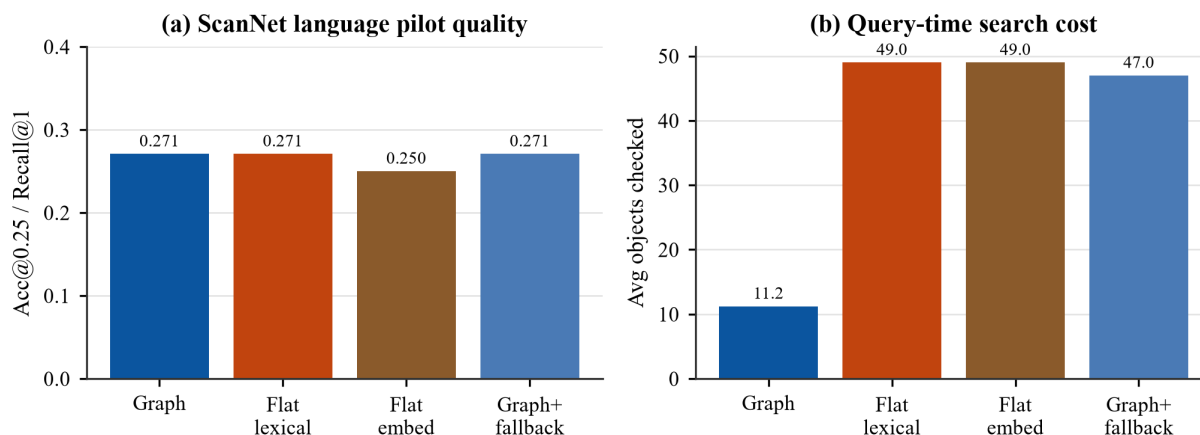
Cumulative estimated context over 150 queries: about 475k flat versus 152k graph.

Official public pilots validate object retrieval over GT maps

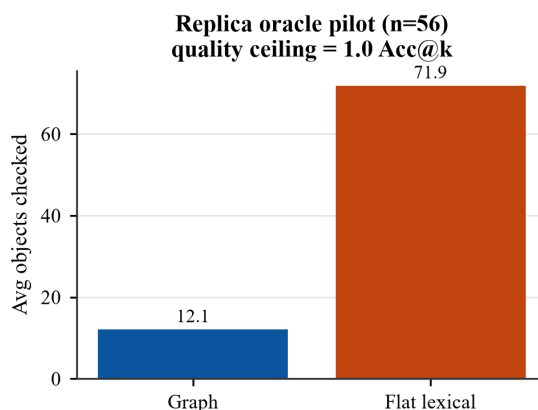
Replica and ScanNet now provide official boxes and language-grounding queries. Graph and flat variants are compared over identical oracle semantic candidates. The key result is reduced candidate inspection at matched flat-lexical quality in these pilots.

Track	Scenes / queries	GT boxes	Graph Acc@0.25	Objects: graph / flat	Interpretation
Replica	8 / 56	575	1.000	12.1 / 71.9	Quality ceiling on oracle candidates
ScanNet	8 / 48	392	0.271	11.2 / 49.0	Matched flat lexical; about 77% fewer objects

BBQ-aligned ScanNet pilot (n=48, oracle candidates)



ScanNet pilot. Graph matches flat lexical Acc@0.25 while checking substantially fewer objects; fallback approaches flat cost.



Replica pilot. Acc@k=1.0 is an oracle-candidate sanity ceiling, not independent perception accuracy.

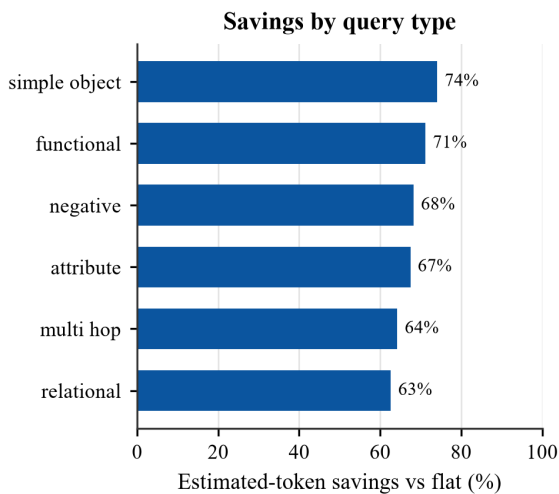
Metric definition: Recall@1 requires the exact target object ID. Acc@0.25 requires predicted and GT 3D boxes to reach IoU ≥ 0.25 . Missing predictions remain in the eligible denominator with IoU zero.

Baselines ran, but fair public-dataset comparison is still missing

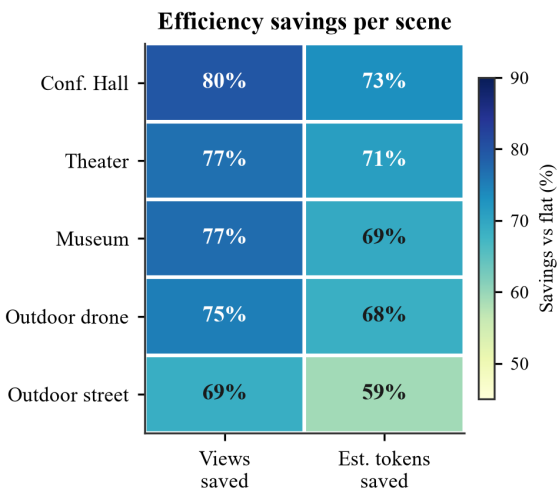
ConceptGraphs now has reproducible native evidence on all five captured scenes. LangSplat executed on its official pretrained sofa asset. These runs prove integration and execution, but not superiority or a fair same-scene public benchmark.

Baseline	Completed evidence	Current boundary	Next fair test
ConceptGraphs	5 captured scenes; 150 canonical outputs	Drone map empty; internal GT; no Replica/ScanNet run	Run selected Replica/ScanNet scenes and identical queries
LangSplat	Official sofa render/query smoke; 1 canonical output	No GT and no captured/public scene comparison	Train/convert one shared public scene with GT
BBQ	Paper studied and metrics/protocol aligned	Official code not reproduced	Same query subset and oracle/perception split
SemanticSplat variants	Graph, fallback, flat lexical, flat embedding	Same-input internal comparisons only	Retain as controlled ablations

Where the current hierarchy saves the most



Savings by query family.



Article readiness

Check	Status	Meaning
ECCV 2026 review format	DONE	Anonymous, line numbered, 10 content pages + references
Claim/evidence consistency	42/42	Includes 97-view count and forbidden stale wording
Related work	13 citations	3DGS semantics, scene graphs, grounding, open-vocabulary maps
Submission portal	PENDING	Paper ID, author profiles/conflicts, upload

What is still missing: science and submission actions

The following items are not hidden blockers; they define exactly what the current paper does not yet prove and what must be completed for submission or a stronger venue.

Gap	Severity now	Why it matters	Completion criterion
Public repository publicity	P0 / policy	ECCV embargo risk because public materials name the intended venue	Make repo private or complete owner-approved remediation
OpenReview submission actions	P0 / blocking	Paper ID, profiles, conflicts, and upload cannot be automated locally	All author profiles complete; ID inserted; PDF uploaded
Oracle semantic candidates	High science	Public pilots do not test automatic perception	Add predicted objects/segments and report non-oracle metrics
Graph quality gap	High science	Internal hit@k trails flat lexical	Calibrated pruning/fallback recovers quality with most savings
Fair external baseline	High science	No same-scene public-dataset BBQ/ConceptGraphs/LangSplat comparison	Identical scenes, queries, GT, and metrics
End-to-end 3DGS evidence	Medium-high	Title frames the system around 3DGS, but evaluation is mainly over semantic views/maps	One reproducible 3DGS reconstruction-to-query experiment
Segmentation metrics	Medium-high	mAcc/mIoU/fmIoU remain N/A without model predictions	Paired prediction and GT arrays with evaluator output
Statistical breadth	Medium	48 ScanNet queries are a pilot, not a broad benchmark	Expanded queries, per-family results, confidence intervals

Immediate policy action: do not push additional venue-labelled changes to the public repository until the team resolves visibility/history. The current report was generated locally and does not change remote visibility.

What the current evidence safely supports

- Hierarchy substantially reduces checked context over the same semantic map.
- Internal lexical retrieval currently pays a measurable hit@k cost.
- Replica and ScanNet oracle-map pilots show candidate-efficiency gains, not semantic-perception accuracy.
- ConceptGraphs and LangSplat execution evidence is real but protocol-limited.
- No superiority claim over BBQ, ConceptGraphs, or LangSplat is currently justified.

Roadmap: submit carefully, then strengthen the science

The fastest credible path is to freeze the workshop submission while starting a parallel research track that removes oracle assumptions and tests the 3DGS claim end to end.

Phase	Window	Work	Exit criteria
1. Submission freeze	15-31 Jul 2026	Resolve publicity; insert paper ID; profiles/conflicts; final human anonymity/citation check; upload	Valid anonymous PDF accepted by portal before deadline
2. Quality recovery	Aug-Sep 2026	Tune branch threshold and fallback; relation diagnostics; expand ScanNet queries; confidence intervals	Graph approaches flat hit@k while retaining most context savings
3. Non-oracle evaluation	Sep-Nov 2026	Predicted segmentation/object proposals; mIoU/mAcc/fmIoU; public-scene ConceptGraphs or BBQ	Same-scene, same-query, same-GT baseline table
4. 3DGS integration	Oct-Dec 2026	Reconstruct/import a 3DGS scene; automatic semantic extraction; query and localization demo	Reproducible reconstruction-to-answer pipeline
5. Main-conference package	Late 2026-2027	Broader datasets, ablations, robustness, failure taxonomy, supplementary artifact	Publication claim supported beyond a workshop-scale pilot

Recommended priority order

- First protect submission eligibility and finish the external OpenReview actions.
- Then recover graph retrieval quality; this is the core algorithmic weakness visible in current evidence.
- Next replace oracle maps with predictions and run one fair public baseline.
- Finally demonstrate the full 3DGS-to-semantic-query path and scale the benchmark.

Final recommendation. The current artifact is suitable as an honest TwinWorld workshop submission after the external submission and publicity actions. A stronger main-conference paper should wait for quality recovery, non-oracle predictions, a fair baseline, and explicit end-to-end 3DGS evidence.

Evidence base

Primary sources: five_scene_graph_vs_flat_v2 metrics; Replica and ScanNet pilot grounding summaries; ConceptGraphs full-run and LangSplat smoke outputs; TwinWorld claim/number/submission audits; papers/twinworld/main.pdf. All provider-token, segmentation, and superiority claims outside these artifacts remain unavailable.